

# STRATUS: Selective Temporal Coupling for Auditable Time-Series Preprocessing

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## Abstract

Temporal recordings contain heterogeneous local quality conditions, yet preprocessing is commonly applied as one global operator sequence. We present STRATUS, an auditable architecture that maps local diagnostics to operational states and explicit actions. Earlier diagnostic and Gaussian-HMM variants show that temporal persistence is not automatically beneficial. We therefore introduce STRATUS-H, a selective hybrid hidden Markov model: complete missing-run length deterministically separates short from long loss, while a discriminative observation model assigns finite samples to Stable, Movement, Unstable-Impulse, or Unstable-Burst substates; a learned Markov transition matrix and Viterbi decoding are applied only within finite blocks. A matched HYBRID-P control removes only the transition terms. On 47 participant-specific reference windows from two public eye-tracking datasets, 33 identifiers are used for training/development and 14 are held out for 2,100 controlled corruptions. STRATUS-H reaches a dataset-macro local-action score of 0.920 and RMSE 14.615, compared with 0.914 and 14.821 for HYBRID-P and 0.882 and 22.703 for the original pointwise diagnostic model. The paired STRATUS-H-minus-HYBRID-P improvement is +0.0057 in local-action score (95% participant-bootstrap interval [0.0043, 0.0071]), with lower MAE and RMSE. The temporal contribution is therefore small but consistent; most of the larger gain over the original diagnostic models comes from the constrained state topology and discriminative observation model. Pointwise-D retains a slightly lower MAE, and an auxiliary shifted-generator check narrows the advantage. The results support selective, explicitly ablated temporal coupling rather than persistence as a default.

## Keywords

time-series preprocessing, data cleaning, operator selection, hidden Markov models, data quality, eye tracking

## 1 Introduction

Temporal data rarely exhibit one homogeneous quality condition. A sensor stream may alternate between reliable observations, genuine rapid dynamics, short interruptions, prolonged loss, and finite artifacts. Nevertheless, preprocessing is commonly expressed as a global pipeline: interpolation, smoothing, clipping, or replacement is configured once and applied throughout a file. The resulting transformation is reproducible, but it silently assumes that the same operator is appropriate at every position.

That assumption is often false. Interpolation can be defensive when a short gap is locally supported, yet it invents values across a long unobserved interval. Smoothing may suppress jitter

but flatten a valid transition. Robust replacement can correct an isolated spike while damaging a legitimate extreme. The central problem is therefore not only how to transform a series, but how to select the locally appropriate action and when to leave data unchanged or missing. From a data-management perspective, this is a local operator-selection problem [1, 14, 22].

Data-cleaning systems select repairs from constraints, statistical evidence, feedback, or downstream objectives [17, 23]. Automated preparation systems such as BoostClean and Diff-Prep search operators or pipelines for predictive tasks [16, 19], while PClean performs probabilistic cleaning from domain models [18]. These systems motivate explicit operator choice, but their decision units and objectives differ from position-specific temporal action plans. Time-series methods exploit temporal context for interpolation, denoising, anomaly handling, and learned imputation [3, 4]; however, temporal coupling can also spread an incorrect local decision. It must therefore be evaluated against an otherwise matched pointwise model.

We present STRATUS, an architecture that separates observable diagnostics, operational state inference, and action execution. Six diagnostics describe missingness, complete gap length, velocity, acceleration, local instability, and plausibility. States denote preprocessing conditions rather than behavioral activities. The policy preserves Stable and Movement samples, interpolates LossShort, leaves LossLong missing, and robustly replaces Unstable samples. This decomposition produces an auditable action path and allows temporal assumptions to be inserted or removed without changing the policy.

The original diagnostic variants reveal an important negative result. POINTWISE-D independently maximizes interpretable diagnostic potentials, whereas STRATUS-D adds a fixed self-transition preference and Viterbi decoding. The fixed persistence terms reduce local-action quality. A diagonal-Gaussian Baum-Welch model, STRATUS-BW, also fails to acquire the short-loss semantics. These outcomes do not show that temporal models are generally harmful; they show that generic persistence and likelihood-based clustering need not align with operational repair decisions.

We therefore introduce STRATUS-H, a selective hybrid HMM. Missing runs are handled by a transparent duration constraint because their complete length directly determines whether interpolation is permissible. Only finite blocks are temporally decoded. A multinomial logistic observation model assigns posterior potentials to Stable, Movement, Unstable-Impulse, and Unstable-Burst substates, and a training-estimated Markov matrix couples adjacent assignments. The two unstable substates share the same rolling-median action. The matched HYBRID-P control uses the identical observation model, features, state topology, and policy but makes independent pointwise decisions. Their contrast isolates the temporal contribution.

We evaluate the model on 47 participant-specific, fully finite eye-tracking windows from ETDD70 and an autism eye-tracking dataset. Thirty-three identifiers form the outer training partition and fourteen remain held out. Hyperparameters are selected only on an inner development subset. The primary test contains ten corruption seeds, three severities, and five corruption settings per held-out participant, yielding 2,100 sequences. An auxiliary shifted-generator evaluation changes positions, durations, movement-label quantiles, jitter distributions, and spike structure.

#### Research Questions.

**RQ1** How can heterogeneous time-series preprocessing be represented as auditable local operator decisions rather than one global pipeline?

**RQ2** Does selective learned temporal coupling improve an otherwise identical discriminative pointwise model, and how does it compare with fixed-persistence and Gaussian-HMM variants?

**RQ3** How do reconstruction error, local action quality, and generator shift alter the conclusions about adaptive preprocessing?

#### Contributions. The paper contributes:

- an auditable diagnostic–state–action representation for local time-series preprocessing;
- a selective hybrid HMM that deterministically constrains missingness states and temporally decodes only ambiguous finite conditions;
- a matched pointwise control that isolates the value of the Markov transition model from the observation model and policy;
- a participant-separated train/development/test protocol with 2,100 held-out corruptions, participant-paired bootstrap intervals, and an auxiliary shifted-generator check;
- a decomposed evaluation combining MAE/RMSE with long-gap safety, short-gap recovery, movement preservation, stable preservation, and bounded unstable-region repair; and
- a public artifact containing code, tests, exact splits, aggregate and sequence-level results, paper sources, and verification scripts.

## 2 Related Work

### 2.1 Data Cleaning and Time-Series Repair

Data cleaning addresses incomplete, inconsistent, implausible, and erroneous data [1, 14, 22]. Because repairs alter data, they must be assessed against explicit quality criteria and intended uses [25]. HoloClean combines constraints, external data, and statistical signals [23], while ActiveClean prioritizes records consequential for model training [17]. Their decision units are primarily records, cells, datasets, or analysis tasks; STRATUS instead selects actions at temporal positions and makes any coupling between adjacent choices explicit.

Time-series cleaning uses temporal order and local context. SCREEN imposes speed constraints [28], while minimum-change repair can use labeled truths [32]. Clean4TSDB combines profiling, detection, and repair [10]. Surveys cover missing, noisy, and inconsistent values [30], while BRITS learns temporal imputations [4]. STRATUS addresses a complementary layer: it selects among heterogeneous actions, including deliberate non-repair, and exposes why each operator is applied.

### 2.2 Eye-Tracking Preprocessing

Eye-tracking data are sensitive to device characteristics, calibration, sampling rate, participant behavior, and task setting [12]. Missing samples can arise from blinks, occlusion, or tracking failure, while coordinate noise and spikes coexist with valid rapid eye movements. Event-detection algorithms consequently depend on noise handling and parameterization. Noise-robust fixation detection and REMoDNaV, for example, address eye-movement classification under noisy measurements [8, 11]. The operational regimes in STRATUS must not be confused with fixation, saccade, or cognitive-state labels. A MOVEMENT regime only states that rapid local dynamics should be preserved.

### 2.3 Hidden-State and Hybrid Sequence Models

HMMs distinguish observed from latent sequences and represent temporal dependence through transitions [21]. Viterbi returns a maximum-score path [29], Baum–Welch estimates generative parameters by expectation maximization [2, 9], and hidden semi-Markov models add explicit duration distributions [31]. Hybrid systems combine discriminative posterior estimates with an HMM transition model so that strong local classification and temporal integration can be ablated separately [24, 27]. Eye-tracking HMMs commonly model fixation patterns, scanpaths, attended regions, or cognitive states [6, 13, 15, 20]. In contrast, STRATUS states denote local preprocessing conditions and directly determine whether values are preserved, interpolated, left missing, or robustly replaced. State semantics and actions remain explicit because a statistically coherent component is not automatically an operational repair state.

## 3 Problem Formulation

Let

$$X_{1:T} = (x_1, \dots, x_T), \quad x_t \in \mathbb{R}^P \cup \{\perp\}, \quad (1)$$

be a recorded time series, where  $\perp$  denotes an unavailable observation. Metadata  $\mathcal{M}$  contain timestamps, sampling information, and a plausibility domain  $\Omega$ . The present realization is offline: complete gap lengths and centered neighborhoods may use observations on both sides of  $t$ .

Let  $\mathcal{O} = \{o^{(1)}, \dots, o^{(K)}\}$  be a library of preprocessing actions. A conventional pipeline applies one fixed composition to the entire series. We instead seek a local action plan

$$\mathbf{a}_{1:T} = (a_1, \dots, a_T), \quad a_t \in \mathcal{O}, \quad (2)$$

that produces  $\widehat{X}_{1:T} = F(X_{1:T}, \mathbf{a}_{1:T}, \mathcal{M})$ , where  $F$  executes the position-specific actions subject to the metadata. Directly selecting  $a_t$  from  $x_t$  is unreliable because the same symptom can imply different actions. We therefore define a finite operational state space  $\mathcal{S}$  and a policy

$$\pi : \mathcal{S} \rightarrow \mathcal{O}, \quad a_t = \pi(z_t). \quad (3)$$

The regime  $z_t$  is inferred from a diagnostic vector

$$d_t = \phi(X_{1:T}, t, \mathcal{M}). \quad (4)$$

With  $D_{1:T} = (d_1, \dots, d_T)$  and  $Z_{1:T} = (z_1, \dots, z_T)$ , the complete processing chain is

$$D_{1:T} \longrightarrow Z_{1:T} \longrightarrow \mathbf{a}_{1:T} \longrightarrow \widehat{X}_{1:T}. \quad (5)$$

A useful solution must satisfy four requirements. Diagnostics and regimes must be interpretable because they control data modification. Any temporal coupling must be explicit and empirically separable from pointwise evidence. The policy must distinguish actions with different risk profiles, particularly supported from

**Table 1: Operational regimes and executed actions.**

| State     | Local condition            | Action         |
|-----------|----------------------------|----------------|
| Stable    | finite and plausible       | preserve       |
| Movement  | genuine rapid change       | preserve       |
| LossShort | gap $\leq 0.5$ s           | interpolate    |
| LossLong  | gap $> 0.5$ s              | keep missing   |
| Unstable  | finite extreme instability | rolling median |

unsupported reconstruction. Finally, evaluation must separate numerical similarity from action appropriateness. We do not search arbitrary pipelines or learn the policy end to end. We assume an explicit operational state space and study whether the appropriate local state can be inferred.

## 4 The STRATUS Model

### 4.1 Architecture and Diagnostic Representation

STRATUS separates diagnostic extraction, regime modeling, optional sequence decoding, and action execution. As summarized in Figure 1, the workflow first transforms the raw time series and metadata into pointwise diagnostics, then infers an operational state path, and finally converts that path into local preprocessing actions through the shared policy  $\pi : \mathcal{S} \rightarrow \mathcal{O}$ . STRATUS-D and STRATUS-BW use persistent sequence decoding; POINTWISE-D is the matched pointwise control. The processed series  $\hat{X}_{1:T}$  is therefore the result of an explicit decision chain rather than one uniform global pipeline.

This separation is deliberate. A threshold exceedance is evidence, not yet a cleaning decision. A large change may represent a valid transition or an artifact, and a missing value can belong to a short or long gap. The regime layer resolves this evidence in temporal context before values are modified.

At time  $t$ , the six-dimensional diagnostic is

$$d_t = [m_t, r_t, c_t, \alpha_t, u_t, p_t]. \quad (6)$$

Here  $m_t$  indicates missingness and  $r_t$  is the sample count of the complete contiguous missing run containing  $t$  (zero outside a run). The first-order diagnostic  $c_t$  is coordinate change divided by elapsed time, while  $\alpha_t$  is the absolute change in  $c_t$  divided by elapsed time. The instability diagnostic  $u_t$  measures distance from a centered seven-sample coordinate median, and  $p_t$  indicates a finite coordinate outside  $\Omega$ . Dynamics are not computed across missing or implausible samples, so individual dynamic components may be undefined even though a diagnostic vector is indexed at every  $t$ ; STRATUS-BW later imputes those components from training statistics. Assigning the complete run length to every missing position prevents the beginning of a long gap from first being classified as short. The sampling-rate estimate converts  $r_t$  to seconds when the policy applies its duration boundary.

### 4.2 Operational States and Policy

The implementation uses

$$\mathcal{S} = \{\text{STABLE}, \text{MOVEMENT}, \text{LOSSSHORT}, \text{LOSSLONG}, \text{UNSTABLE}\}. \quad (7)$$

These are operational states rather than behavioral labels. Table 1 defines the common policy.

Stable-region smoothing is disabled in all reported experiments. For LOSSSHORT, linear interpolation is applied only to samples assigned to that state. LOSSLONG positions remain missing. UNSTABLE positions are replaced by a centered seven-sample rolling median.

### 4.3 Sequence Inference and Selective Hybrid HMM

The model family separates three questions: what local evidence is available, whether neighboring assignments should interact, and which action follows from a state. Figure 1 summarizes the resulting architecture. Missingness is not delegated to a learned finite-state classifier. Complete run length gives a hard, auditable split between LOSSSHORT and LOSSLONG. Finite positions are instead assigned among four internal states,

$$\mathcal{F} = \{\text{STABLE}, \text{MOVEMENT}, \text{UNSTABLEIMPULSE}, \text{UNSTABLEBURST}\}. \quad (8)$$

The two unstable substates are distinct during inference but both map to the public UNSTABLE action. Splitting isolated spikes from sustained jitter prevents one duration pattern from dominating a single broad instability state.

*Diagnostic baselines.* The original POINTWISE-D and STRATUS-D variants use explicit compatibility scores  $e_t(s)$ . Finite samples receive Stable evidence; high velocity provides Movement evidence; extreme jitter, acceleration, or plausibility violations provide Unstable evidence; and complete gap duration provides loss-state evidence. POINTWISE-D selects  $\arg \max_s e_t(s)$  independently. STRATUS-D adds a self-transition bonus of 1.25, a switch penalty of  $-0.75$ , and Viterbi decoding. Their difference is therefore a matched test of fixed persistence. STRATUS-BW remains a five-component diagonal-Gaussian HMM trained by Baum-Welch with post-hoc semantic alignment on training labels. It tests whether unsupervised likelihood components acquire the operational state semantics.

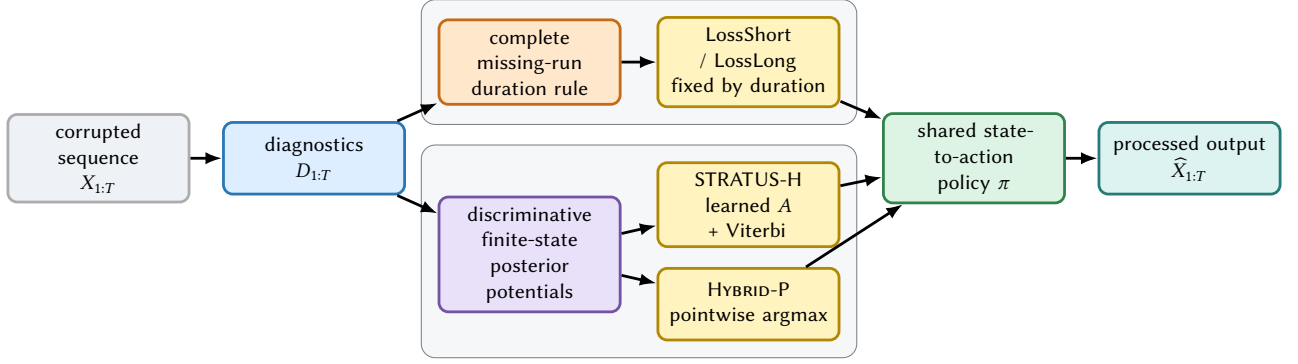
*Discriminative observation potentials.* For finite sample  $t$ , a feature vector  $f_t$  contains log-transformed velocity, acceleration, and local instability; within-sequence ranks of these three diagnostics; centered short-window means; longer-window maxima; deviations from short-window context; and the plausibility indicator. The window durations are expressed in seconds and converted to samples using the recording rate. A multinomial logistic model estimates

$$q_\theta(s | f_t), \quad s \in \mathcal{F}. \quad (9)$$

It is trained only from corruptions generated for outer-training participants. Class-balanced sampling and class weighting prevent the dominant Stable class from suppressing the two instability patterns. The regularization parameter  $C \in \{0.1, 1, 10\}$  is chosen on development participants.

*Selective Markov decoding.* Transition counts between finite training labels estimate a matrix  $A$  with additive 0.5 smoothing; the initial distribution  $\rho$  is estimated analogously. For every maximal finite block  $b = [u, v]$ , STRATUS-H decodes

$$Z_b^* = \arg \max_{Z_b} \left[ \sum_{t=u}^v \log q_\theta(z_t | f_t) + \lambda \log \rho_{z_u} + \lambda \sum_{t=u+1}^v \log A_{z_{t-1}, z_t} \right]. \quad (10)$$



**Figure 1: Selective temporal coupling in STRATUS. Missing runs follow a deterministic duration constraint. The matched finite-state variants share features, posterior potentials, state topology, and policy; only STRATUS-H adds a learned transition matrix and Viterbi decoding.**

**Table 2: Model and policy parameters. “Dev.” denotes participant-separated development selection.**

| Parameter                     | Value       | Source          |
|-------------------------------|-------------|-----------------|
| Short/long gap boundary       | 0.5 s       | fixed policy    |
| Unstable repair window        | 7 samples   | fixed policy    |
| Movement proxy quantile       | 0.90        | fixed generator |
| Finite feature windows        | 0.05/0.15 s | fixed design    |
| Logistic $C$                  | 1           | dev. grid       |
| Transition strength $\lambda$ | 1           | dev. grid       |
| Transition pseudocount        | 0.5         | fixed design    |
| Training corruption seeds     | 0–4         | fixed protocol  |
| Test corruption seeds         | 0–9         | fixed protocol  |

The transition strength  $\lambda \in \{0, 0.1, 0.25, 0.5, 0.75, 1\}$  is selected on development participants. At  $\lambda = 0$ , Equation (10) reduces exactly to HYBRID-P. With  $\lambda > 0$ , posterior state potentials are integrated by an HMM transition model, following the hybrid posterior/HMM pattern used in earlier sequence classifiers [24, 27]. We call the model hybrid rather than a fully generative HMM because its local terms are discriminative posterior potentials rather than a learned density  $p(f_t | z_t)$ .

Hard constraints prohibit finite samples from entering loss states and prohibit missing samples from entering finite states. Finite blocks are decoded independently, so transitions cannot bridge missing intervals. Both unstable substates are mapped to UNSTABLE before policy execution. Thus the HMM is used only where temporal context might distinguish genuine movement, isolated spikes, sustained jitter, and stable observations.

#### 4.4 Parameterization, Auditability, and Complexity

All variants expose the predicted state and resulting action before values are modified. Table 2 distinguishes fixed policy choices from development-selected parameters. The 0.5-second boundary and seven-sample repair window are policy choices retained from the original experiment. Observation regularization and transition strength are selected on eight development identifiers drawn only from the outer training partition; the selected values are  $C = 1$  and  $\lambda = 1$ . The final hybrid model is then refitted on all 33 outer-training identifiers before the 14 test identifiers are processed.

The intermediate path also supports failure localization. A changed value can be traced to its diagnostics, finite or missing branch, assigned state, and policy action. Let  $M_f = 4$  be the number of finite substates and  $q = 13$  the number of finite features. Logistic inference is  $O(TqM_f)$ ; pointwise assignment is  $O(TM_f)$  after scoring; and Viterbi is  $O(TM_f^2)$  with  $O(TM_f)$  backpointers. Complete-run labeling, diagnostics, and action execution are linear for fixed windows. The small state space keeps the temporal layer inexpensive relative to feature extraction and model fitting.

## 5 Eye-Tracking Case Study

### 5.1 Data and Domain Instantiation

Eye tracking exposes the conflicts that motivate local operator selection. Trackers produce bounded two-dimensional coordinates, often for both eyes, but samples may disappear during blinks, occlusion, or loss of pupil detection. At the same time, genuine saccadic motion creates rapid coordinate changes that can resemble spikes or jitter. Sampling frequencies and missing-value encodings also differ between devices and exports [8, 11, 12]. We therefore instantiate the generic observation  $x_t$  as a fused binocular gaze coordinate and derive every diagnostic from timestamps and coordinate geometry rather than from clinical labels or fixation/saccade annotations.

Table 3 maps the generic diagnostic vector to the case-study realization. Left and right coordinates are averaged whenever at least one eye is valid. Diagnostics involving temporal differences are computed only across adjacent finite and plausible samples. This prevents the boundary of a gap from creating an artificial velocity spike. The state names remain operational: MOVEMENT means preserve rapid dynamics, not that a particular physiological eye-movement event has been classified.

**Datasets and harmonization.** We use ETDD70 and the Eye-Tracking Autism dataset. ETDD70 contains binocular recordings of 70 Czech children during three reading tasks and was recorded at approximately 250 Hz [26]. The autism dataset distributes experiments across 25 CSV exports; one export may contain several participants [7]. Clinical labels are not used. The datasets provide heterogeneous real trajectories on which controlled quality states can be injected.

To bound the runtime of the controlled stress test, the loader requests the first ten lexicographically sorted CSV files from each local dataset folder. This deterministic subset is fixed before model fitting and is not selected using labels or experimental

**Table 3: Eye-tracking realization of the generic STRATUS diagnostics.**

| Symbol     | Generic quantity       | Eye-tracking realization                                                                                    | Operational role                             |
|------------|------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| $m_t$      | missingness            | at least one fused gaze coordinate unavailable                                                              | identifies tracking loss                     |
| $r_t$      | missing-run length     | sample count of the complete contiguous invalid run; converted to seconds using the estimated sampling rate | separates recoverable and unsupported gaps   |
| $c_t$      | first-order change     | Euclidean gaze displacement divided by elapsed time                                                         | provides evidence for genuine rapid movement |
| $\alpha_t$ | second-order change    | absolute change in $c_t$ divided by elapsed time                                                            | exposes abrupt, non-sustained dynamics       |
| $u_t$      | local instability      | distance from a centered seven-sample coordinate median                                                     | captures jitter and isolated spikes          |
| $p_t$      | plausibility violation | finite coordinate outside the empirical dataset domain $\Omega$                                             | flags geometrically implausible observations |

**Table 4: Loaded data and participant-separated reference windows.**

| Dataset | Files | IDs | Usable | Train/Test | Hz     |
|---------|-------|-----|--------|------------|--------|
| ETDD70  | 10    | 10  | 10     | 7/3        | 250.00 |
| Autism  | 9     | 58  | 37     | 26/11      | 59.88  |

outcomes. All ten ETDD70 files load successfully. For the autism data, one file lacks the required left-eye point-of-regard columns and is excluded, leaving nine files. Left and right valid gaze coordinates are averaged. Zeros and the autism missing marker “-” are treated as unavailable. ETDD70 timestamps are converted from microseconds; autism timestamps are normalized from milliseconds within each participant-trial-stimulus recording. Table 4 summarizes the resulting data.

Coordinate plausibility bounds are estimated separately from finite raw values as the 99.9th percentile rounded upward to 100 coordinate units. This yields domains of  $1600 \times 1000$  for ETDD70 and  $1400 \times 1200$  for Autism. These are empirical coordinate domains, not physical screen-resolution claims.

## 5.2 Reference Extraction and Participant-Safe Model Selection

Candidate windows are formed within one participant and one recording. A run is broken by invalid coordinates, non-increasing timestamps, or a timestamp increment exceeding three times the recording median. The first continuous window spanning at least eight seconds is retained per recording; one candidate is then selected per participant with seed 42. All selected windows are fully finite. The median durations are 8.003 seconds for ETDD70 and 8.016 seconds for Autism.

Participants are split separately by dataset into 33 outer-training and 14 outer-test identifiers. ETDD70 contributes seven training and three test identifiers; Autism contributes 26 and 11. No identifier occurs in both partitions. Eight of the outer-training identifiers form an inner development partition (two ETDD70 and six Autism), leaving 25 for inner model fitting. The development partition selects  $C$  and  $\lambda$ ; it is not reused for final test evaluation. After selection, the hybrid observation model and transition counts are refitted on all outer-training identifiers.

For each training reference window, the matched generator creates five corruption seeds, three severities, and five corruption settings. At most 120 sequences per dataset enter model fitting. The classifier uses injected operational labels only from training participants. This supervision is a substantive difference from STRATUS-BW, whose Gaussian parameters are learned without labels and whose labels are used only for component alignment.

The distinction is explicit: STRATUS-H asks whether a semantically constrained, supervised finite-state model can use temporal context once the operational label space has been defined.

The outer test partition is never used for threshold choice, logistic fitting, transition estimation, or semantic mapping. The empirical coordinate domains are carried as dataset metadata from the fixed v7 extraction artifact. Raw third-party data and coordinate-level reference windows are not redistributed. The repository instead commits the deterministic extraction code, extraction summary, participant split, metadata, reference-file SHA-256 record, and aggregate and sequence-level experimental outputs.

## 6 Experimental Design

### 6.1 Matched and Shifted Stress Tests

The primary matched-generator test retains the original controlled protocol. Every held-out window is degraded under ten random seeds, three severity multipliers  $\{1.0, 1.7, 2.5\}$ , and five settings: short gap, long gap, jitter, spikes, and their mixture. Short gaps occupy 2.5% of samples times severity and long gaps 8% times severity; jitter is Gaussian noise over an 8% interval; and spikes replace 1.5% of finite samples times severity by large deviations. Natural Movement reference labels are assigned to the upper decile of clean point-to-point displacement before injected states overwrite overlaps. These are *injected operational reference labels*, not expert-annotated physiological truth. The primary design yields  $14 \times 10 \times 3 \times 5 = 2,100$  test sequences.

The auxiliary shifted-generator check deliberately changes the mechanisms available during training. Positions are randomized; gap durations are sampled continuously on either side of the 0.5-second policy boundary; the Movement proxy uses the upper 15% rather than 10% of clean displacements; jitter combines Laplace noise with low-frequency drift; and spikes are heavy-tailed clusters of one to three samples. Two independent seeds, three severities, five settings, and the same 14 held-out participants yield 420 additional sequences. This check is smaller than the primary test and is reported as sensitivity evidence, not a second confirmatory benchmark.

### 6.2 Compared Methods

The central comparison contains five levels of temporal and semantic structure:

- **Pointwise-D** and **STRATUS-D**: interpretable diagnostic potentials without and with fixed Viterbi persistence;
- **STRATUS-BW**: a diagonal-Gaussian Baum-Welch HMM with training-only semantic alignment;

**Table 5: Participant-separated evaluation design.**

| Item                                   | Value   |
|----------------------------------------|---------|
| Reference windows                      | 47      |
| Outer train / test IDs                 | 33 / 14 |
| Inner fit / development IDs            | 25 / 8  |
| Primary test sequences                 | 2,100   |
| Shifted-generator sequences            | 420     |
| Primary bootstrap replicates           | 2,000   |
| Observation $C$ / transition $\lambda$ | 1/1     |

- **Hybrid-P**: the new supervised finite-state observation model with hard gap constraints and no transitions;
- **STRATUS-H**: the identical hybrid model with learned transitions and Viterbi decoding.

The shared policy preserves Stable and Movement, interpolates LossShort, leaves LossLong missing, and replaces both unstable substates with a centered seven-sample median. We retain the oracle, short-gap-only, and global baselines from the v7 artifact to provide policy and failure-mode context. The decisive HMM ablation is STRATUS-H versus HYBRID-P because every component except temporal coupling is identical.

### 6.3 Metrics, Aggregation, and Inference

Let  $X^{\text{ref}}$ ,  $X^{\text{deg}}$ , and  $\hat{X}$  denote the clean reference, degraded input, and processed output. MAE and RMSE are computed from Euclidean coordinate error on finite reference/output pairs [5]; retention reports the finite-output fraction. Action-specific metrics are:

$$H_L = \Pr(\hat{x}_t \neq \perp \mid z_t^{\text{ref}} = \text{LossLong}), \quad (11)$$

$$R_S = \Pr(\hat{x}_t \neq \perp \mid z_t^{\text{ref}} = \text{LossShort}), \quad (12)$$

movement preservation  $P_M$ , stable preservation  $P_S$ , and unstable repair gain  $G_U$ . For unstable positions,  $G_U$  is the relative reduction in error from degraded input to output; missing outputs receive no improvement credit. The bounded component

$$G_U^+ = \text{clip}(G_U, 0, 1) \quad (13)$$

is the value displayed in result tables and used in the composite, resolving the distinction between raw and clipped repair gain. The local-action score is

$$Q_{\text{local}} = \frac{(1 - \bar{H}_L) + \bar{R}_S + \bar{P}_M + \bar{P}_S + \bar{G}_U^+}{5}. \quad (14)$$

Components are first averaged over applicable corruption cases per participant. Participants are then averaged within each dataset, and ETDD70 and Autism receive equal weight in the primary dataset-macro result. Undefined components are omitted only before participant aggregation. We report every component with  $Q_{\text{local}}$  rather than treating it as a universal quality index.

Paired participant bootstrap intervals resample identifiers within each dataset and preserve method pairing; 2,000 replicates are used. The primary contrast is STRATUS-H minus HYBRID-P. A positive  $Q_{\text{local}}$  interval and negative error intervals support an independent temporal contribution. Weight sensitivity recomputes the score under equal, safety-heavy, repair-heavy, no-Stable, and finite-only component weights.

## 7 Results

### 7.1 Primary Matched-Generator Results

Table 6 reports the dataset-macro primary results. STRATUS-H has the highest deployable local-action score, 0.920, and the lowest RMSE, 14.615. POINTWISE-D retains the lowest MAE, 3.566, narrowly below STRATUS-H at 3.598. Both hybrid variants preserve complete short-gap recovery and zero long-gap filling. Their main advantage is stronger bounded unstable repair:  $G_U^+ = 0.775$  for STRATUS-H and 0.771 for HYBRID-P, compared with 0.523 for POINTWISE-D.

The fair temporal contrast is STRATUS-H minus HYBRID-P. Its  $Q_{\text{local}}$  difference is +0.0057 with 95% participant-bootstrap interval [0.0043, 0.0071]. MAE changes by  $-0.227$  ( $[-0.320, -0.148]$ ) and RMSE by  $-0.206$  ( $[-0.348, -0.063]$ ), where negative is favorable. All fourteen test identifiers have a higher participant-level  $Q_{\text{local}}$  under STRATUS-H. Temporal decoding therefore contributes independently, but the effect is small. The much larger difference between the hybrid family and POINTWISE-D cannot be attributed to the HMM; it arises primarily from the supervised finite-state observation model, expanded unstable topology, and hard missingness constraints.

Against POINTWISE-D, STRATUS-H improves  $Q_{\text{local}}$  by 0.0383 ( $[0.0299, 0.0468]$ ) and RMSE by 8.088 ( $[6.479, 9.733]$  lower). The MAE difference is only +0.031 with an interval spanning zero. Thus the methods emphasize different error structures: POINTWISE-D preserves a larger fraction of Stable and Movement samples exactly, while STRATUS-H removes large finite artifacts more aggressively and thereby lowers squared error.

Figure 2 makes the matched temporal contrast concrete. It shows two held-out mixed-corruption sequences for which the difference between HYBRID-P and STRATUS-H is visually legible. These examples were selected only for illustration after evaluation; they were not used for tuning or statistical inference. Both methods share the same diagnostics, classifier, hard gap rules, and action policy. Their differing state strips therefore expose the effect of sequence decoding itself. The pointwise model produces short Stable/Movement/Unstable fluctuations, whereas STRATUS-H suppresses many isolated switches and yields more coherent unstable intervals. The repaired trajectories remain similar because temporal coupling changes only a minority of positions, consistent with the small aggregate effect.

### 7.2 Dataset Differences

The result is not uniform across datasets. On ETDD70, STRATUS-H reaches  $Q_{\text{local}} = 0.963$ , MAE 0.922, and RMSE 4.020, clearly ahead of POINTWISE-D. On Autism, it reaches the highest  $Q_{\text{local}}$  at 0.877, but POINTWISE-D has lower MAE (4.764 versus 6.274). In both datasets, however, STRATUS-H improves over the matched HYBRID-P control, showing that the temporal contribution is not generated by only one source.

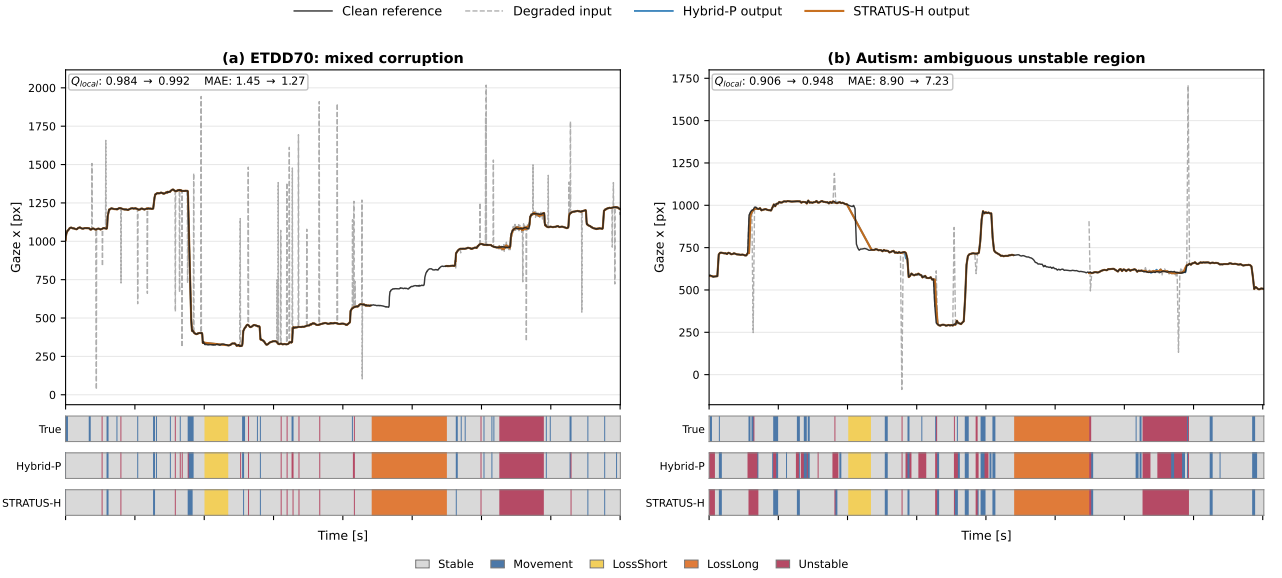
The Autism result prevents a simplistic “hybrid wins” claim. Its stronger unstable repair comes with more changes to finite samples and a higher mean coordinate error. The decomposition is therefore essential: local action quality, MAE, and RMSE answer different questions.

### 7.3 Shifted Generator and Weight Sensitivity

Figure 3 compares the primary generator with the auxiliary shifted mechanisms. Under shift, STRATUS-H remains the highest equal-weight method at  $Q_{\text{local}} = 0.907$ , followed by HYBRID-P

**Table 6: Dataset-macro primary results.**  $G_U^+$  is the bounded repair component actually used in  $Q_{\text{local}}$ . **Bold values mark the best deployable method; retention is descriptive.**

| Method               | MAE↓         | RMSE↓         | Ret.  | $H_L$ ↓      | $R_S$ ↑      | $P_M$ ↑      | $P_S$ ↑      | $G_U^+$ ↑    | $Q_{\text{local}}$ ↑ |
|----------------------|--------------|---------------|-------|--------------|--------------|--------------|--------------|--------------|----------------------|
| Oracle policy        | 1.526        | 7.048         | 0.945 | 0.000        | 1.000        | 0.994        | 1.000        | 0.784        | 0.956                |
| STRATUS-H            | 3.598        | <b>14.615</b> | 0.945 | <b>0.000</b> | <b>1.000</b> | 0.881        | 0.944        | <b>0.775</b> | <b>0.920</b>         |
| HYBRID-P             | 3.825        | 14.821        | 0.945 | <b>0.000</b> | <b>1.000</b> | 0.872        | 0.928        | 0.771        | 0.914                |
| POINTWISE-D          | <b>3.566</b> | 22.703        | 0.945 | <b>0.000</b> | <b>1.000</b> | <b>0.904</b> | <b>0.981</b> | 0.523        | 0.882                |
| STRATUS-D            | 4.849        | 32.456        | 0.945 | <b>0.000</b> | <b>1.000</b> | 0.874        | 0.981        | 0.350        | 0.841                |
| STRATUS-BW           | 5.542        | 39.121        | 0.928 | <b>0.000</b> | 0.000        | 0.852        | 0.973        | 0.387        | 0.642                |
| Short-gap only       | 10.408       | 58.273        | 0.945 | <b>0.000</b> | <b>1.000</b> | 0.807        | 1.000        | 0.000        | 0.761                |
| Global interpolation | 13.865       | 64.097        | 1.000 | 1.000        | <b>1.000</b> | 0.807        | 1.000        | 0.000        | 0.561                |
| Global smoothing     | 19.095       | 38.258        | 0.935 | 0.074        | 0.236        | 0.489        | 0.000        | 0.669        | 0.464                |



**Figure 2: Qualitative examples of selective temporal coupling on held-out mixed-corruption sequences.** Each panel shows the clean reference, degraded input, and repaired outputs for one coordinate, followed by the injected regime path and the predictions of HYBRID-P and STRATUS-H. The ETDD70 example changes  $Q_{\text{local}}$  from 0.984 to 0.992 and MAE from 1.45 to 1.27; the Autism example changes  $Q_{\text{local}}$  from 0.906 to 0.948 and MAE from 8.90 to 7.23. Both models share the local observation model and policy, so the reduction in short state fluctuations visualizes the selective HMM contribution. The examples are illustrative; all quantitative claims use the complete held-out evaluation.

**Table 7: Per-dataset comparison of the new hybrid family and the original pointwise diagnostic model.**

| Dataset | Method      | MAE↓         | RMSE↓        | $Q_{\text{local}}$ ↑ |
|---------|-------------|--------------|--------------|----------------------|
| Autism  | STRATUS-H   | 6.274        | 25.211       | <b>0.877</b>         |
| Autism  | HYBRID-P    | 6.715        | 25.603       | 0.867                |
| Autism  | POINTWISE-D | <b>4.764</b> | 28.747       | 0.876                |
| ETDD70  | STRATUS-H   | <b>0.922</b> | <b>4.020</b> | <b>0.963</b>         |
| ETDD70  | HYBRID-P    | 0.934        | 4.039        | 0.961                |
| ETDD70  | POINTWISE-D | 2.368        | 16.660       | 0.888                |

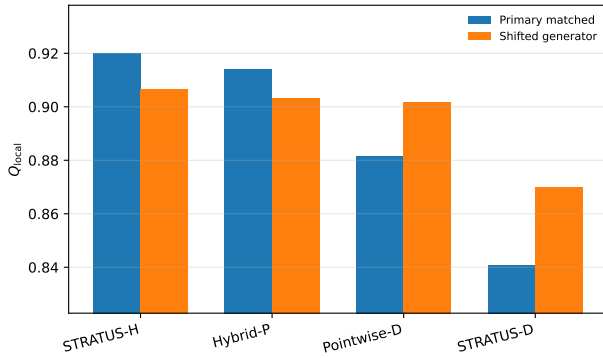
at 0.903 and POINTWISE-D at 0.902. The isolated STRATUS-H minus-HYBRID-P difference remains positive at +0.0035 with interval [0.0015, 0.0054]; MAE improves by 0.215 ([0.130, 0.314] lower), while the RMSE interval narrowly crosses zero. The contrast with POINTWISE-D is no longer decisive in  $Q_{\text{local}}$ , and POINTWISE-D has the lower MAE. The shifted test therefore

supports the small temporal effect but weakens the broader architecture claim, as expected under a generator not seen during training.

The primary ranking STRATUS-H > HYBRID-P > POINTWISE-D > STRATUS-D is unchanged under all five reported component-weight schemes, including omission of Stable preservation and a finite-only score. Under the shifted generator, STRATUS-H ranks first in four schemes; repair-heavy weighting gives HYBRID-P a small advantage (0.813 versus 0.811). Hence the headline result is not an artifact of exact Stable preservation, but the narrow temporal advantage can depend on how unstable repair is weighted after distribution shift.

## 7.4 Answering the Research Questions

**RQ1.** The diagnostic-state-action decomposition represents pre-processing as an inspectable local plan. Hard constraints can



**Figure 3: Dataset-macro  $Q_{\text{local}}$  under the primary matched generator and the auxiliary shifted generator. The shifted check changes positions, durations, label quantile, jitter distribution, and spike structure.**

encode actions whose permissibility is directly observable, while learned inference is reserved for ambiguous finite conditions.

**RQ2.** Generic persistence is not sufficient: fixed Viterbi terms and the Gaussian Baum–Welch model remain weak. Selective learned transitions improve the matched discriminative pointwise control by a small but participant-consistent amount. The benefit is therefore conditional on the observation semantics and constrained topology.

**RQ3.** No single scalar captures the outcome. STRATUS-H has the best local-action score and RMSE, while POINTWISE-D retains the best primary MAE and remains competitive under generator shift. Reporting the five action components prevents either error metric or  $Q_{\text{local}}$  from hiding these trade-offs.

## 8 Discussion

### 8.1 Implications for Adaptive Data Preparation

The central methodological lesson is not that an HMM should replace pointwise preprocessing. It is that temporal coupling should be inserted only after the operational state space and policy are explicit, and then tested against a matched no-transition control. In the original diagnostic model, a generic preference for persistence absorbs finite artifacts into neighboring runs and hurts. In the hybrid model, the observation semantics are stronger, gap states are removed from the ambiguous search space, and transitions are learned from training labels; under those conditions, temporal decoding yields a small consistent improvement.

The result also separates architectural gain from HMM gain. Moving from POINTWISE-D to the hybrid family changes features, supervision, and the internal instability topology. Moving from HYBRID-P to STRATUS-H changes only temporal coupling. The first difference is large; the second is modest. This distinction is important for data-management systems because an impressive end-to-end improvement can otherwise be misattributed to the sequence model rather than to a better representation or policy constraint.

Selective non-repair remains a first-class action. All principal variants recover short losses while leaving long losses unavailable, avoiding the misleading retention gain of global interpolation. The same pattern can generalize to other domains when action preconditions can be expressed explicitly: for example, bounded interpolation horizons, sensor-specific plausibility

ranges, or minimum support for replacement. The policy should remain auditable even when the observation model becomes more complex.

### 8.2 Threats to Validity and Limitations

**Injected operational labels.** The study does not use expert annotations of natural eye-tracking artifacts. LossShort and LossLong are generated on opposite sides of the same 0.5-second policy boundary used by the model. Their perfect recovery is therefore principally a definition and implementation check. Movement is a displacement-quantile proxy, not a physiological event label. The shifted test decouples its quantile and samples durations around the boundary, but it remains synthetic.

**Supervision and generator coupling.** STRATUS-H is trained on injected labels and is not an unsupervised discovery model. The primary training and test corruptions share a generator family, which favors learning its finite-artifact patterns. The shifted test changes several mechanisms and preserves the matched HMM advantage, but uses only two seeds and the same reference participants. It does not substitute for naturally annotated errors or a third domain.

**Sample size and split.** The study contains 47 usable identifiers and only three held-out ETDD70 identifiers. One fixed outer split supports exact paired comparisons but does not quantify split-to-split variability. Participant bootstrap intervals describe uncertainty conditional on this split. Repeated participant splits and leave-one-participant-out analyses remain necessary for stronger population claims.

**Offline features.** Complete gap length and centered windows use future context. The present realization is unsuitable for strictly causal streaming without delayed decisions, online duration estimates, or an explicit-duration model.

**Model interpretation.** The local logistic terms are posterior potentials, so STRATUS-H is a hybrid posterior/HMM decoder rather than a fully generative HMM. Its transition matrix is estimated from operational training labels. The classifier uses class balancing, and probability calibration is not separately assessed. A linear-chain conditional model, calibrated posterior-to-likelihood conversion, mixture emissions, or hidden semi-Markov durations may provide a more principled temporal layer.

**Metrics and policy.** MAE/RMSE are conditional on finite outputs and must be read with retention and gap safety.  $P_M$  summarizes mean displacement and may hide compensating local errors;  $P_S$  uses exact preservation; and  $Q_{\text{local}}$  is a study-specific equal-weight composite. Weight sensitivity is reassuring but not universal. The seven-sample median also limits the oracle, so state inference alone cannot achieve perfect repair.

**Comparison scope.** BoostClean, DiffPrep, and PClean address different data models or require downstream labels/domain programs, so they are discussed rather than executed as drop-in time-position baselines. The experimental baselines isolate temporal, gap, and global-transformation behavior but do not exhaust modern automated preparation systems.

### 8.3 Future Work

The immediate next step is external validation: naturally annotated tracking failures, an independent eye-movement detector or manual Movement subset, a third sensor domain, and repeated participant splits. Methodologically, duration-aware or conditional sequence models should be compared with the same

pointwise observation potentials. Calibration and transition uncertainty should be reported across multiple initializations. For causal operation, centered diagnostics must be replaced by online counterparts and latency included in evaluation. Finally, policy learning is promising only if hard safety constraints and action provenance remain visible.

## 9 Conclusion

STRATUS represents adaptive time-series preprocessing as diagnostics, operational states, and explicit local actions. The revised selective hybrid model uses deterministic duration constraints for missingness and learned temporal coupling only for ambiguous finite conditions. On the participant-separated primary stress test, STRATUS-H achieves the highest local-action score (0.920) and lowest RMSE (14.615), while the original POINTWISE-D retains a slightly lower MAE. Against the identical HYBRID-P control, the learned Markov layer provides a small but stable improvement in action quality and error. The main end-to-end gain nevertheless comes from the constrained topology and discriminative observation model rather than from temporal decoding alone. Under a shifted corruption generator, the isolated temporal effect persists but the difference from POINTWISE-D narrows. The broader conclusion is therefore methodological: temporal assumptions in data preparation should be selective, auditable, and justified by matched out-of-sample ablations.

## 10 Artifacts

The implementation and experimental artifact are available at <https://github.com/eyetracking-data/STRATUS/tree/main/stratus-regime-aware-preprocessing>. The repository contains reusable modules, the primary and shifted-generator runners, the exact participant split and extraction metadata, tests for hard gap constraints and matched decoding, aggregate and sequence-level results, paper sources, and verification scripts. Third-party raw data and coordinate-level reference windows are not redistributed; original sources, licenses, expected directory layouts, and the hash of the local v7 reference file are documented.

After the public source datasets have been placed locally, the deterministic extraction command recreates the reference windows; the central experiment command then generates POINTWISE-D, STRATUS-D, HYBRID-P, and STRATUS-H. The result verifier checks row counts, selected hyperparameters, safety invariants, hashes, and headline values without refitting. The paper package records the artifact version and the files required for the primary table and bootstrap contrasts.

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